

In this lecture notes we look at finding an inverse of an invertible matrix and some more logic.

Inverse of an invertible matrix

Let A be an $n \times n$ invertible matrix, from the invertibility theorem we have that the identity matrix I_n and A are row equivalent and so there exists elementary matrices $E_1, E_2, \dots, E_k$ such that $I_n = E_k \dots E_2 E_1 A$ and since A is invertible we also have that $A^{-1} = E_k \dots E_2 E_1 = E_k \dots E_2 E_1 I_n$. This last equation says that the exact same row operations used to reduce A into I_n , will at the same time transform I_n into A^{-1} .

Therefore when finding inverse of A , it will be useful to create an $n \times 2n$ matrix.

$$\left(\begin{array}{cccc|cccc} a_{11} & a_{12} & \cdots & a_{1n} & 1 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2n} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & 0 & 0 & \cdots & 1 \end{array} \right)$$

and row reduce A to I , thus obtaining $[A|I_n] \xrightarrow{\text{row operations}} [I_n|A^{-1}]$.

Example We find an inverse of matrix $A = \begin{pmatrix} 1 & -3 & 2 \\ 0 & 1 & -1 \\ -1 & 0 & -1 \end{pmatrix}$ and associated elementary matrices.

$$\begin{aligned} \left(\begin{array}{ccc|ccc} 1 & -3 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 & 0 & 1 \end{array} \right) &\xrightarrow{R_3 \rightarrow R_3 + R_1} \left(\begin{array}{ccc|ccc} 1 & -3 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & -3 & 1 & 1 & 0 & 1 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow R_3 + 3R_2} \left(\begin{array}{ccc|ccc} 1 & -3 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 & 3 & 1 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow -\frac{1}{2}R_3} \left(\begin{array}{ccc|ccc} 1 & -3 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1/2 & -3/2 & -1/2 \end{array} \right) \\ &\xrightarrow{R_2 \rightarrow R_2 + R_3} \left(\begin{array}{ccc|ccc} 1 & -3 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1/2 & -1/2 & -1/2 \\ 0 & 0 & 1 & -1/2 & -3/2 & -1/2 \end{array} \right) \\ &\xrightarrow{R_1 \rightarrow R_1 - 2R_3} \left(\begin{array}{ccc|ccc} 1 & -3 & 0 & 2 & 3 & 1 \\ 0 & 1 & 0 & -1/2 & -1/2 & -1/2 \\ 0 & 0 & 1 & -1/2 & -3/2 & -1/2 \end{array} \right) \\ &\xrightarrow{R_1 \rightarrow R_1 + 3R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 3/2 & -1/2 \\ 0 & 1 & 0 & -1/2 & -1/2 & -1/2 \\ 0 & 0 & 1 & -1/2 & -3/2 & -1/2 \end{array} \right) \end{aligned}$$

and so $A^{-1} = \begin{pmatrix} 1/2 & 3/2 & -1/2 \\ -1/2 & -1/2 & -1/2 \\ -1/2 & -3/2 & -1/2 \end{pmatrix}$.

We have used a finite sequence of operations (in respective order) $e_1 : R_3 \rightarrow R_3 + R_1$, $e_2 : R_3 \rightarrow R_3 + 3R_2$, $e_3 : R_3 \rightarrow -\frac{1}{2}R_3$, $e_4 : R_2 \rightarrow R_2 + R_3$, $e_5 : R_1 \rightarrow R_1 - 2R_3$, and $e_6 : R_1 \rightarrow R_1 + 3R_2$.

Their associated elementary matrices are:

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}, E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1/2 \end{pmatrix}, E_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$E_5 = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_6 = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We then have that $E_6 \cdots E_2 E_1 A = I_n$ and so if we are to write A as a product of elementary matrices we would have $A = E_1^{-1} E_2^{-1} \cdots E_6^{-1}$.

$$E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, E_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \text{ (write down inverses of all the above elementary matrices).}$$

Notice that finding inverses of elementary matrices is an easy task, the inverse of elementary matrix E associated to operation e is the elementary matrix associated to the operation (of the same type as e) that has inverse effect to e . That is, on operations, inverse of $R_i \leftrightarrow R_j$ is $R_j \leftrightarrow R_i$, inverse of $R_i \rightarrow \lambda R_i$ ($\lambda \neq 0$) is $R_i \rightarrow \frac{1}{\lambda} R_i$ and inverse of $R_i \rightarrow R_i + \lambda R_j$ is $R_i \rightarrow R_i - \lambda R_j$.

Some more logic

Negation of ‘conjunction’ and ‘disjunction’ of statements (De Morgan’s laws): Given statements P and Q , we have that $\neg(P \wedge Q)$ is logically equivalent to $\neg P \vee \neg Q$ and $\neg(P \vee Q)$ is logically equivalent to $\neg P \wedge \neg Q$. Below we show one logical equivalence and leave the other as an exercise.

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$	$\neg(P \wedge Q) \iff \neg P \vee \neg Q$
1	1	0	0	1	0	0	1
1	0	0	1	0	1	1	1
0	1	1	0	0	1	1	1
0	0	1	1	0	1	1	1

Example Contrapositive of statement ‘For every $n \times n$ matrix A , if $A^2 = A$ and $A \neq I_n$, then A is not invertible’ is ‘For every $n \times n$ matrix A , if A is invertible, then $A^2 \neq A$ or $A = I_n$.’

In proving the above contrapositive statement we may use logical equivalence $P \implies Q \vee R \iff P \wedge \neg Q \implies R$. That is suppose A is invertible and $A^2 = A$, then we have $A^{-1}(A^2) = A^{-1}(A)$, that is, $A = I_n$.

Negation of ‘quantifiers’: Given a statement P , $\neg(\forall_x P(x))$ is logically equivalent to $\exists_x(\neg P(x))$ and $\neg(\exists_x P(x))$ is logically equivalent to $\forall_x(\neg P(x))$.

Example The negation of statement ‘for all real number x , there exists a natural number n such that $x < n$ ’ is statement ‘there exists a real number x , such that for all natural numbers n , $x \geq n$ ’. With our usual real numbers and natural numbers, the first statement is true, while its negation is false.